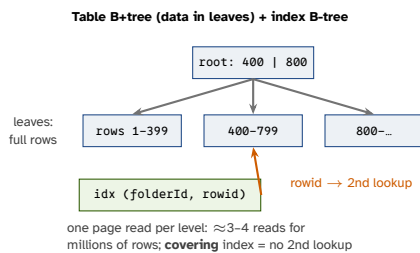
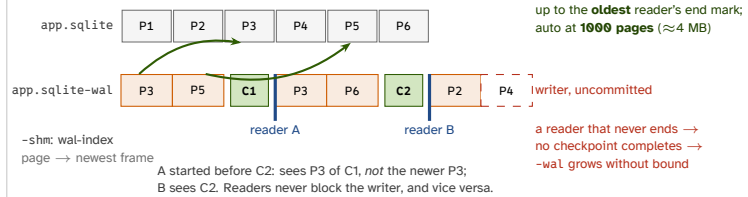


**In one line:** SQLite is a library, not a server: one **file** of fixed-size **pages** (4096 B default) holding **B-trees** — one per table (rows by rowid) and one per index. Many connections may **read**, only **one may write**. In **WAL** mode commits are *appended* to `-wal`, readers keep a snapshot (*end mark*), and a **checkpoint** copies pages back. Core Data, SwiftData, GRDB and Android Room all sit on it.



#### WAL mode — append, snapshot, checkpoint



#### How it works

- Pages:** 512-65 536 B, 4096 default (since 3.12.0). A table is a **B-tree** keyed by the 64-bit rowid (INTEGER PRIMARY KEY aliases it) — data only in leaves. An index is a B-tree of (columns, rowid); WITHOUT ROWID tables are index B-trees.
- Rollback journal** (DELETE, the default): original pages copied to `-journal`, the db written *in place*; commit = delete the journal. A writer committing locks readers out.
- WAL** (PRAGMA journal\_mode=WAL, **persistent**): commit = append a commit frame to `-wal`. Readers use `-shm` (shared memory) — so **same host only**, no network file systems. Big win for apps: UI reads never wait for a background import.
- Transactions:** BEGIN = **DEFERRED** — no lock until first access; a read that later writes must *upgrade*, and fails with **SQLITE\_BUSY** if another connection wrote meanwhile. BEGIN IMMEDIATE takes the write lock up front (= EXCLUSIVE in WAL). A read transaction is a **snapshot**: it never sees later commits.
- SQLITE\_BUSY = another *connection* holds the lock (busy timeout = wait and retry). No timeout set → fails at once. The real fix: **one writer** in the whole app.
- Durability:** WAL + synchronous=NORMAL (recommended) syncs only at checkpoint — a commit may be lost on *power loss*, never corrupted. FULL syncs every commit.
- Off by default: PRAGMA foreign\_keys (per connection). GRDB turns it on.

#### GRDB — one writer, many readers

- One** DatabaseQueue or DatabasePool per file, for the app's lifetime. **Queue** = one connection, everything serialised. **Pool** = WAL, a serial **writer** + concurrent **readers**, each read isolated.
- Every write {} and read {} is a transaction; writes are serialised (so no SQLITE\_BUSY inside one process); no writes inside read; non-reentrant. Escape hatches are named `...WithoutTransaction / unsafe...`
- ValueObservation.tracking { db in ... }: records the *region* it read; any commit touching it → re-fetch. Only committed changes; may **coalesce**; removeDuplicates(); .values(in:) async sequence or start(in:onError:onChange:).
- DatabaseMigrator: named migrations, each in its own transaction, applied list stored in the db; FK checks deferred to commit. **Never edit a shipped migration**. eraseDatabaseOnSchemaChange is for development only.

#### Example — pool, migration, write, observe

```
var cfg = Configuration()
cfg.busyMode = .timeout(5) // app-group peers: wait, don't fail
let db = try DatabasePool(path: url.path, configuration: cfg)
var m = DatabaseMigrator()
m.registerMigration("v1") { db in
    try db.create(table: "note") { t in
        t.autoIncrementedPrimaryKey("id")
        t.column("folderId", .integer).notNull()
        t.column("title", .text).notNull()
        t.column("updatedAt", .datetime).notNull()
    }
    try db.execute(sql: """
        CREATE INDEX idx_note_folder ON note(folderId, updatedAt, title)""")
    try m.migrate(db) // each migration = one transaction
    try await db.write { db in try Note(folderId: 1, title: "Hi").insert(db) }
    let obs = ValueObservation.tracking { db in // region: note table
        try Note.filter(Column("folderId") == 1).fetchAll(db)
    }
    for try await notes in obs.values(in: db) { render(notes) }
```

- Tests: DatabaseQueue() with no path = in-memory db — run the *real* migrator on it, plus golden db files written by old app versions.

#### Query plans, indexes, N+1

```
EXPLAIN QUERY PLAN SELECT title FROM note
WHERE folderId = ? ORDER BY updatedAt;
-- no index: SCAN note + USE TEMP B-TREE FOR ORDER BY
-- (folderId): SEARCH note USING INDEX ... (folderId=?)
-- + USE TEMP B-TREE FOR ORDER BY
-- (folderId, updatedAt, title):
-- SEARCH note USING COVERING INDEX idx_note_folder (folderId=?)
```

- SCAN** = every row; **SEARCH** = B-tree seek; **COVERING** = no table lookup. Order: **equality** columns first, then the ORDER BY/range one. TEMP B-TREE = a sort you could index away. Every index costs each write.
- N+1**: 1 query for folders + 1 per folder. Fix: JOIN, IN (...), or GRDB including(all:) (one query per association, not per row).
- FTS5**: USING fts5(title, body) + MATCH + ORDER BY rank (bm25); GRDB FTS5() + synchronize(withTable:). Needs SQLITE\_ENABLE\_FTS5 (GRDB docs: custom build). LIKE '%x%' is a scan.

#### On iOS — files, locks, backups

- Three files:** db + `-wal` + `-shm`. Copy, move, back up or protect only the db → lost commits. Put them in **one directory** and treat it as a unit.
- Data Protection:** default Class C (UntilFirstUserAuthentication). `.complete (A)`: key gone ~10 s after lock → background writes fail (disk I/O error).
- `0xdead10cc`: suspended holding a file lock in a **shared container** (app group) → killed. GRDB: observesSuspensionNotifications + Database.suspendNotification.
- Backup** live: backup API (GRDB backup(to:)) or VACUUM INTO 'copy.sqlite' (consistent, compacted). Never cp mid-transaction.
- Migrations** without a library: PRAGMA user\_version (int at header offset 60) — read, run steps above it in one transaction, write it back.
- Core Data** SQLite stores default to WAL since iOS 7; **SwiftData** is Core Data underneath. **Android**: Room = SQLite + compile-time-checked DAOs, Flow observation.

#### Traps

- ! Two pools on one file — two writers, SQLITE\_BUSY, observations miss commits.
- ! Long read transaction (a paused cursor, a huge export) → checkpoint starvation, `-wal` of gigabytes, slower reads.
- ! Read-then-write in a DEFERRED transaction → SQLITE\_BUSY at the upgrade; write transactions from the start (GRDB write).
- ! "Add an index" without EXPLAIN QUERY PLAN before and after.
- ! Deleting or overwriting the db but not its `-wal`, or unlinking a file still open — a mispaired log is a documented corruption path. So is linking **two SQLite copies** (system + a bundled one): one's `close()` can drop the other's POSIX locks.

#### Remember

**Pages** → **B-trees** → **one writer**. WAL: *append, snapshot, checkpoint* — and the oldest reader decides when the log can shrink.